SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING - CSA17 ARTIFICIAL INTELLIGENCE LAB

Day 1 - Program 10	Course Outcome: CO4	Language: Python 3

Question

Write the python program to implement A* algorithm

Aim

To implement the A* informed search algorithm to find the least-cost path from a start node to a goal node.

Algorithm / Explanation

A* keeps an open list ordered by f(n) = g(n) + h(n) using a priority queue (heapq). g(n) is the cost from the start and h(n) is an admissible heuristic estimate to the goal.

With an admissible heuristic A* is complete and returns an optimal path.

Program

```
# A* Algorithm on a weighted graph
import heapq

graph = {
    'S': {'A': 1, 'B': 4},
    'A': {'B': 2, 'C': 5, 'D': 12},
    'B': {'C': 2},
    'C': {'D': 3, 'G': 7},
    'D': {'G': 2},
    'G': {}
}
h = {'S': 7, 'A': 6, 'B': 4, 'C': 4, 'D': 2, 'G': 0}   # admissible heuristic

def a_star(start, goal):
    open_list = [(h[start], 0, start, [start])]          # (f, g, node, path)
    closed = {}
    while open_list:
        f, g, node, path = heapq.heappop(open_list)
        if node == goal:
            return path, g
        if node in closed and closed[node] <= g:
            continue
        closed[node] = g
        for nb, cost in graph[node].items():
            ng = g + cost
            heapq.heappush(open_list, (ng + h[nb], ng, nb, path + [nb]))
    return None, float('inf')

path, cost = a_star('S', 'G')
print("Path found :", " -> ".join(path))
print("Total cost :", cost)
```

Output

```
Path found : S -> A -> B -> C -> D -> G
Total cost : 10
```

Result

Thus the Python program was executed successfully and the output was verified.